



BUSINESS BLUEPRINT

AOP AUTOMATION FOR DMD HDPE PLANT

Honeywell

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Business Blueprint

AOP Automation for DMD HDPE Plant

Prepared for:

Reliance Industries Ltd

Submitted by:

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Revision History

Rev.	Date	Originated By	Description of status and changes
0.0	01-09-2024	HON	Issued for Review
1.0	16-09-2024	HON	Issued for Review (revised norms logic)
2.0	10-02-2025	HON	Issued for Review (updated RIL comments)
3.0	20-02-2025	HON	Issued for Review (Resolved RIL comments)
4.0	25-04-2025	HON	Issued for Approval
5.0	18-07-2025	HON	Site Specific Changes done and logic updated. Issued for approval
6.0	18-08-2025	HON	Norms Logic and Shutdown allocation logic update.

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1. Introduction

The purpose of this document is to capture the detailed specifications for the Digitalized Annual Operating Plan (AOP) for HDPE plant for DMD site.

This document is based on the Technical Proposal submitted by Honeywell to Reliance. It also refers to data collected during the workshop conducted at RCP with various Polyethylene plant users across RIL sites.

1.1 Executive Summary

Reliance Industries Limited is India's leading private sector organization operating in various business verticals. The manufacturing sector is one such business vertical. RIL owns and operates multiple polyethylene plants across various sites.

Defining and finalizing the Annual Operating Plan is a critical annual business process carried out by different functional teams. At present this activity is carried out manually.

A Rev-0 budget is usually completed in Nov / Dec for the succeeding Financial Year (FY) and reviewed. The comments are addressed with updated information from the Business as the next revised budget by the end of Jan. Further revisions (if any) are incorporated in Feb / March to reflect updated business input as applicable.

The entries from the final rev of the budget are finalized in BPC before the beginning of the succeeding FY.

RIL aims to automate the AOP preparation process using a Data-Driven Model. This model leverages the last five years of AOP data from MIIS and incorporates plant-specific business rules (such as plant availability, running hours, feedstock availability, seasonal impact on production, etc.) to derive AOP figures in the expected formats.

As part of digitalization, Honeywell will support building a platform that will automate the Annual Operations Planning process for PE technology.

1.2 Reference Documents

Document ID	Revision	Title	Document
O-836243B24R01	R01	AOP Automation for O2C Proposal	 O-836243B24R01 RIL AOP Automation f
PE effective Operating Hours calculation	-	RIL AOP - PE Maintenance Screen – Reliance to Confirm on logic	

DMD HDPE DCT	-	Digital AOP DMD PE DCT	 PE DCT_DMD.xlsx
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Table 1 Reference Documents

2. AOP Design Basis

The below table captures different functions for the scope of AOP digitization design –

Sr. No.	Functions	Covered in Document
1.	Health, Safety and Environment (HSE)	For later release post review and discussion
2.	Process/Operations	Covered
3.	Reliability and Maintenance	For later release post review and discussion

Table 2 AOP Functional Scope

Function-wise detailed design basis is captured in the below sections.

2.1 Process and Operations Function

In this section, the budgeted Production and Norms for a plant are derived, using a standard approach as briefed in the sections below for the following criteria

- Budget Production/Production AOP
- Norms/Consumption AOP

Steps to generate the Production and Consumption AOP are captured in the below flow diagram.

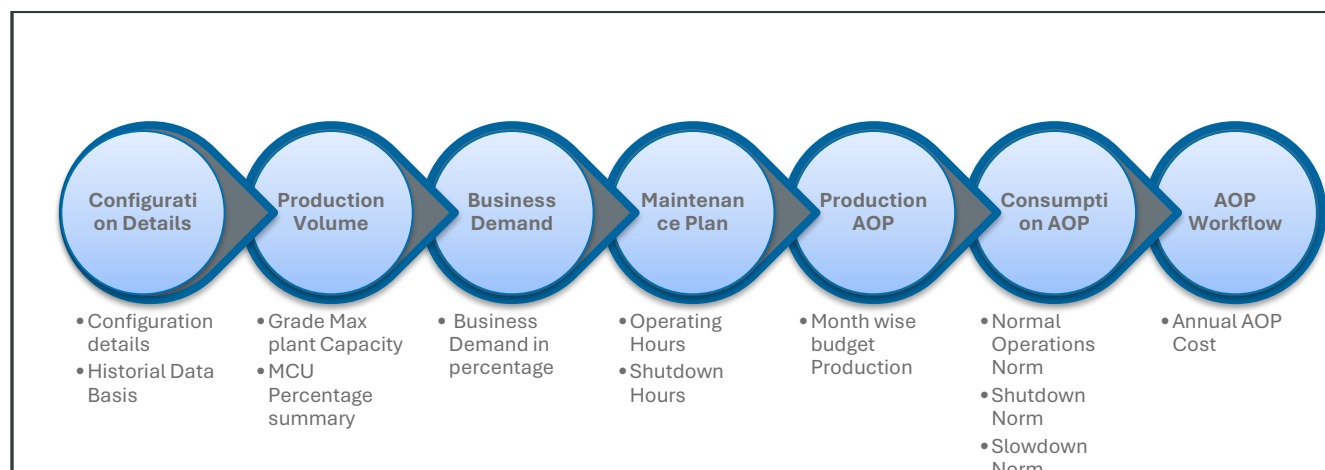


Figure 1 Digitized AOP Flow Diagram

2.1.1 Budget Production

Below are the steps captured to derive the budgeted AOP production values.

2.1.1.1 Configuration

The configuration screen serves as the foundation for the digital AOP process, allowing users to input essential data for production and consumption AOP calculations.

Plant-specific data are required to calculate Production and consumption. data, Startup losses for raw material & Utilities, other losses. There are different tabs available to configure the above parameter under the Configuration screen to support Production and Consumption Norms calculations and reports generation.

By default, the values manual entries entered for the current year will be automatically carried forward to the next year.

In configuration screen, below additional features and tabs are provided.

- **AOP historical period** basis added where Start and End date is provided to select historical period to calculate MCU and Norms data for selected period by default 05 years data will be loaded. AOP Design basis will reflect the selected period as AOP basis, users can add additional basis points for summary.
- **import and export** options to import and export manual entry parameters from configuration screen is provided.
- **Startup losses** are material wise losses during the startup period.
- **Other losses** are material wise other losses.
- **Shutdown Rate (Per HR)** is rate of losses during shutdown period.

The screenshot below highlights the details captured in the digitized AOP platform
Historical Period Basis.

Production & Norms Basis

Year: 2025-26
Vertical: PE
Site: DMD
Plant: HDPE

1 Production & Norms Basis
2 Shutdown/TA Activities
3 Slowdown Activities
4 Production Target
5 Business Demand
6 Maintenance Details
7 Month wise Production plan
8 Steady State Consumption (Norm/Quantity)
9 Shutdown Consumption (Quantity)
10 Slowdown Consumption (Norm/Quantity)
11 Overall AOP Consumption (Norm/Quantity)
12 Feed Stock Availability

PE / DMD / HDPE

AOP Historical Period Basis

Start Date: 01-04-2024
End Date: 31-10-2024
Load
(Last refreshed data on: 10-07-2025 10:04 AM for the period from 01-04-2024 to 31-10-2024)

AOP Design Basis

Manual Input configuration screens

❖ Startup Losses

Startup Losses Other Losses Shutdown Rate (Per Hr) Receipe															
Particulars	UOM	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Jan	Feb	Mar	Remark	
Raw Material															
Ethylene	MT	10	8	6	4	8	10	12	14	6	8	12	11	Test	
Utility Consumption															
LDPE Cooling Water	KM3	12	14	16	18	20	22	24	11	10	15	15	11	Test	
RAW WATER	KM3	16	17	22	21	13	15	12	14	16	15	25	23	Test	
Ret steam condensate	M3	23	14	25	14	18	17	12	21	19	20	21	10	Test	
D M Water	M3	18	21	18	23	16	12	14	10	13	17	19	20	Test	
AIR(INST+PLANT)	NM3	24	22	20	18	16	14	12	10	13	15	17	18	Test	
Power_Dis	MWH	16	23	24	17	18	25	21	19	21	14	33	17	Test	
Gaseous Nitrogen	NM3	10	19	18	17	16	15	14	13	32	28	18	19	Test	
HP Steam_Dis	MT	17	20	18	16	13	12	20	27	28	22	16	15	Test	

❖ Other Losses

❖ Shutdown Rate

Startup Losses	Other Losses	Shutdown Rate (Per Hr)	Receipts	Export	Import	Save
Particulars	:	Values	:	Remark	:	
▼ Raw Material						
Ethylene		0.003		ok		
▼ Utility Consumption						
Power_Dis		0.378		test		
AIR(INST+PLANT)		189.14		ok		
LDPE Cooling Water		0.053		ok		
D M Water		0.2761972		ok		
Ret steam condensate		0.276		ok		
RAW WATER		0.006		ok		
HP Steam_Dis		0.028		ok		
Gaseous Nitrogen		0.168		ok		

Figure 2 Digitized AOP screen for Configuration/manual Inputs

2.1.1.2 Production Target

For example: For grade M60075 and if AOP basis period is selected as April-24 to Oct-24. The last MC value recorded for M60075 in MCU portal that might be 31st Oct 23 will be selected as Basis for next year production calculation by converting it to TPH.

With this logic All grade MC is derived for DMD HDPE

NOTE- AOP historical period basis added in configuration screen where user can change Start date

and End date to change the AOP design basis, accordingly MCU and Norms historical data will be considered for calculation.

The following logic has been formulated to derive MC values.

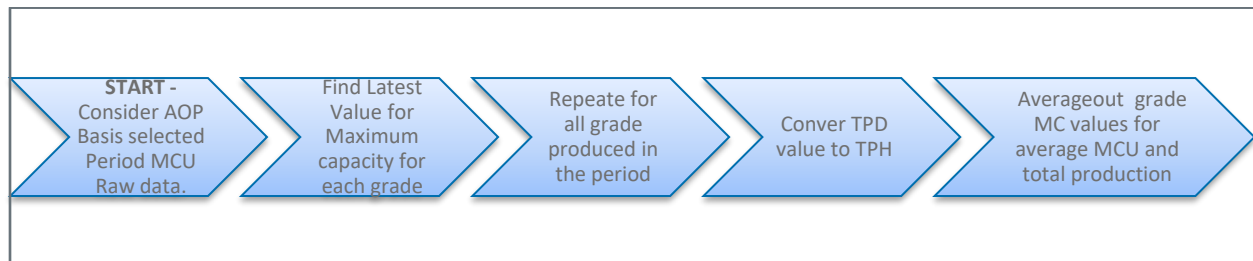


Figure 3 Digitized AOP Steps to derive MC values for DMD HDPE

If the user needs to modify the production rate this can be done by providing a specific reason.

This information is captured in the AOP digitization portal as seen in the screenshot below.

</

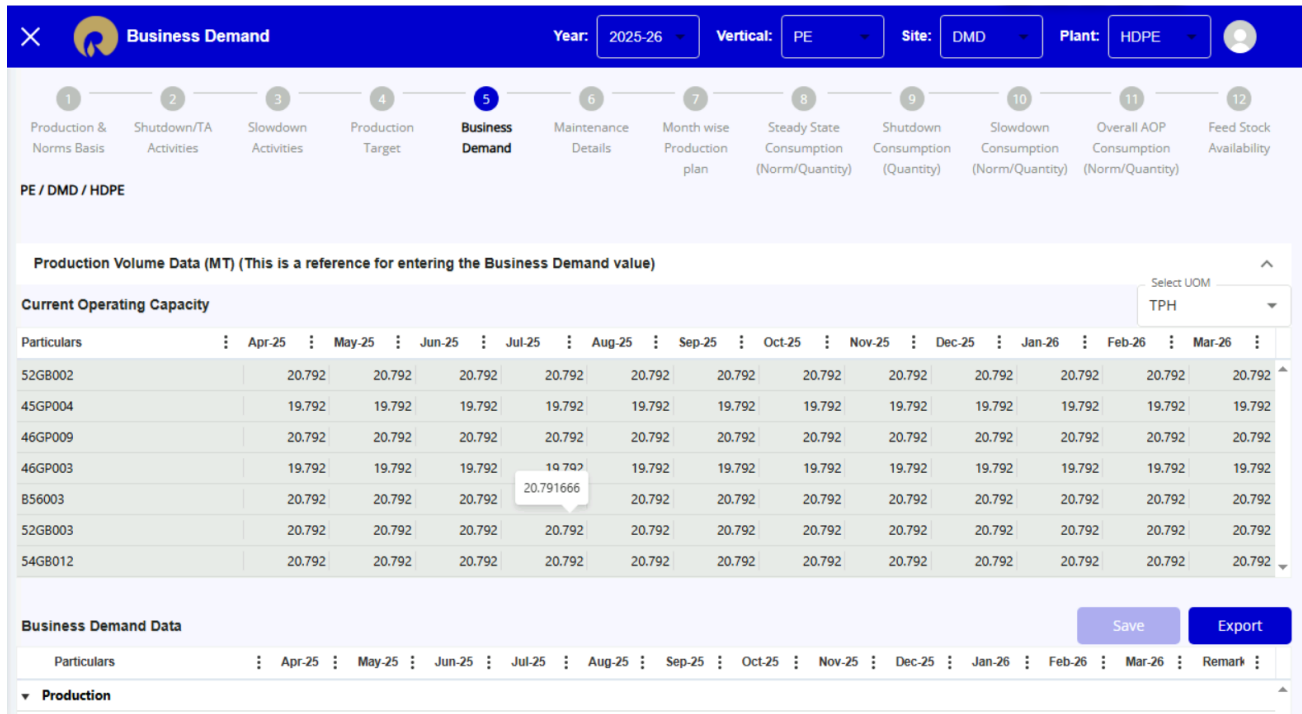
Figure 4 Digitized AOP screen for Production Target

The '**Calculate**' button on this screen allows the user to compute the MC value if any manual changes have been made to the AOP basis selection on the configuration screen

2.1.1.3 Business Demand

The grade mix demand will be provided by the Business team and added in AOP portal manually. In the AOP digitization application, Percent monthly demand for each individual grade will be recorded.

An example of the same is mentioned in the table below.



Business Demand Year: 2025-26 Vertical: PE Site: DMD Plant: HDPE

1 Production & Norms Basis 2 Shutdown/TA Activities 3 Slowdown Activities 4 Production Target 5 **Business Demand** 6 Maintenance Details 7 Month wise Production plan 8 Steady State Consumption (Norm/Quantity) 9 Shutdown Consumption (Quantity) 10 Slowdown Consumption (Norm/Quantity) 11 Overall AOP Consumption (Norm/Quantity) 12 Feed Stock Availability

PE / DMD / HDPE

Production Volume Data (MT) (This is a reference for entering the Business Demand value)

Current Operating Capacity Select UOM TPH

Particulars	Apr-25	May-25	Jun-25	Jul-25	Aug-25	Sep-25	Oct-25	Nov-25	Dec-25	Jan-26	Feb-26	Mar-26
52GB002	20.792	20.792	20.792	20.792	20.792	20.792	20.792	20.792	20.792	20.792	20.792	20.792
45GP004	19.792	19.792	19.792	19.792	19.792	19.792	19.792	19.792	19.792	19.792	19.792	19.792
46GP009	20.792	20.792	20.792	20.792	20.792	20.792	20.792	20.792	20.792	20.792	20.792	20.792
46GP003	19.792	19.792	19.792	20.791666	19.792	19.792	19.792	19.792	19.792	19.792	19.792	19.792
B56003	20.792	20.792	20.792	20.792	20.792	20.792	20.792	20.792	20.792	20.792	20.792	20.792
52GB003	20.792	20.792	20.792	20.792	20.792	20.792	20.792	20.792	20.792	20.792	20.792	20.792
54GB012	20.792	20.792	20.792	20.792	20.792	20.792	20.792	20.792	20.792	20.792	20.792	20.792

Business Demand Data Save Export

Particulars	Apr-25	May-25	Jun-25	Jul-25	Aug-25	Sep-25	Oct-25	Nov-25	Dec-25	Jan-26	Feb-26	Mar-26	Remark
Production													

Figure 5 Grade-wise Business demand

2.1.1.4 Maintenance/Shutdown

The operations team will provide the shutdown and slowdown plan to calculate month-wise operating hours available for production.

The screenshot below depicts the shutdown activities planned, as will be captured in the AOP digitized platform.

The user has to enter the SD-From and SD-To date and time to calculate the duration in hours, or SD-From and Duration (hrs) to calculate the SD-To date/time.

Figure 6 Digitized AOP screen for Shutdown activities

The slowdown activities can be calculated based on the slowdown rate and hours. This is called Basis Shutdown, equivalent of Slowdown.

$$\text{Basis shutdown equivalent of Slowdown hrs} = (\text{Slowdown rate} * \text{Slowdown hrs}) / \text{MCU}$$

The below screen depicts slowdown activities planned as will be captured in the digitized platform.

Figure 7 Digitized AOP screen for Slowdown Activities

Using this number, the operating hours available in a month will be calculated as -

$$\text{Number of Operating Hours in a month} = \text{Total Number of hrs in a month} - (\text{Shutdown Hrs} + \text{Slowdown converted to equivalent SD Hrs})$$

The screenshot below captures an example of the number of operating hours, Shutdown hours, and effective slowdown loads available in a month wise.

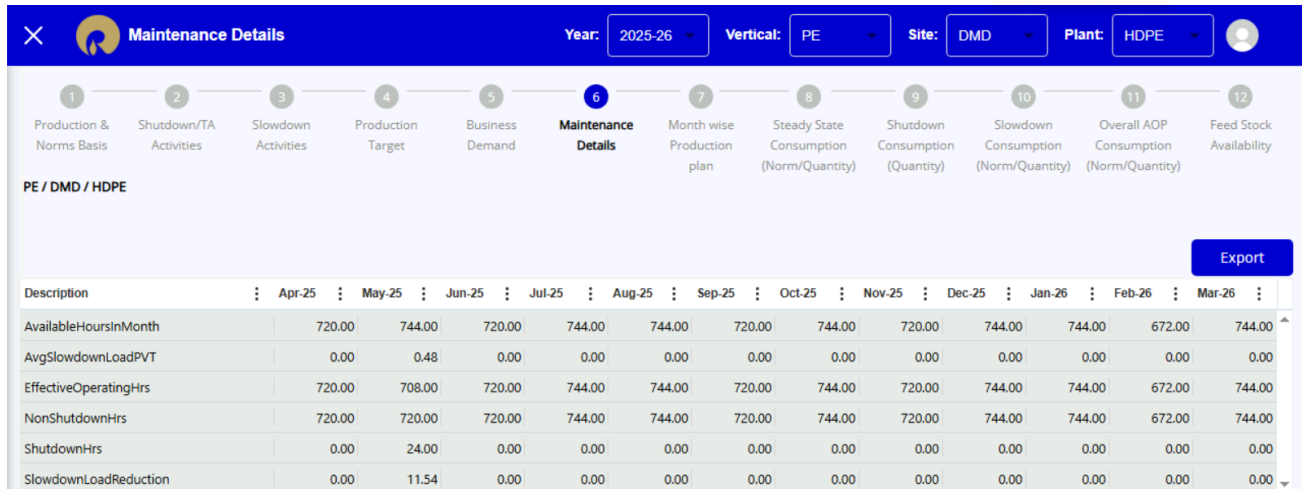


Figure 8 Number of operating hours

Note: RIL to validate slowdown and shut down hours calculated in System and raise any exception observed.

2.1.1.5 Calendar Year Based Production Calculations

Using the Production Volume data, business demand, and maintenance plan, the month-wise and grade-wise budgeted production quantity will be derived.

The calculation logic is as below.

For a single-grade

$$\begin{aligned}
 \text{Total Budgeted Monthly Production} \\
 &= \text{Average MCU of All grades captured in section 2.1.1.2 (TPH)} \\
 &\quad * \text{Operating hours in a month (hr)}
 \end{aligned}$$

$$\begin{aligned}
 \text{Each Grade Production for a month} &= (\text{Total Budgeted Monthly Production} * \\
 &\quad \text{Percentage of Grade production demand (\%)}) / 100
 \end{aligned}$$

Calendar based production calculation explained above will be used to calculate monthly production

Hence, Month – wise production would be,

The screenshot below captures the calculated budgeted production values.

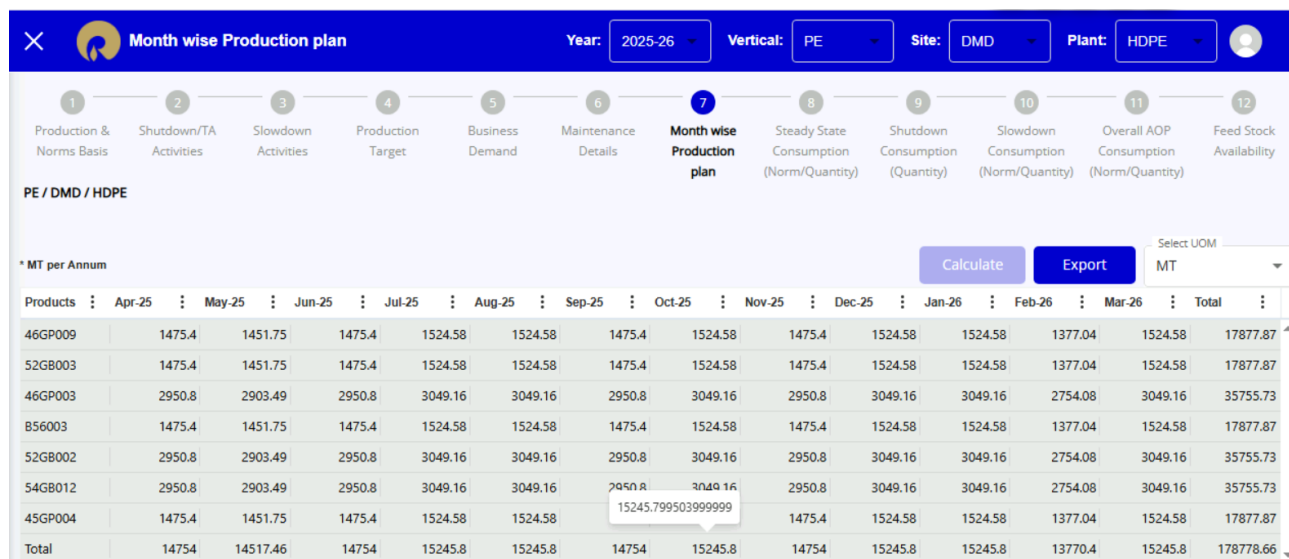


Figure 9 Grade wise Budget Production

2.1.2 Norms

Below are the steps captured to derive the Norms value.

2.1.2.1 Steady-state Operations

At a high level, the norms calculation are categorized as below –

1. Best Achieved –

Raw material, byproduct & utilities norms are generally covered under this calculation basis. Based on the Selected basis AOP Period corresponding to the MCU derived in section 2.1.1.2. High and Low range of MCU (+/- 10% of MCU) is calculated. Based on derived high and low range of production MIIS data for selected period is filtered for consecutive days production. Consecutive days' selection is preferably 5 days, or if not available, 4, 3, 2, or 1 day. the days in which production lies within the high and low range and have consecutive days production are selected for norms calculation. Norms are calculated for each material and each day from the filtered data. respective material norms are then averaged out for Final norm value.

Below is the formulated logic for deriving the Best Achieved Norms

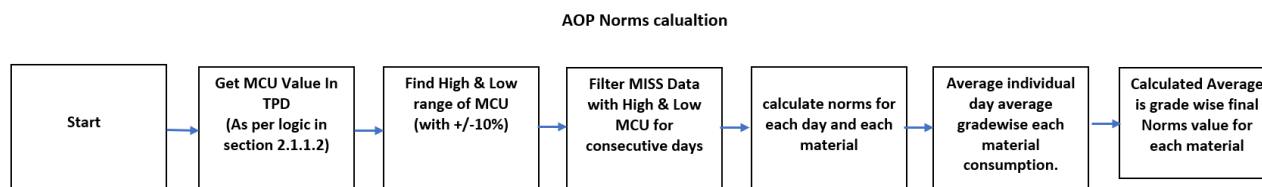


Figure 10 Norm Calculation Flow-Best Achieved

2. SAP Process order basis –

Catalyst-chemicals norms are generally covered under this calculation basis. Based on the Selected basis AOP Period, SAP data for each process order is filter for catalyst & chemicals for each grade. Process order wise consumption aggregation is calculated. For the same process order Production is aggregated to calculate final production number for the process order. Further each material norm is calculated for each process order using the formula below.

$$\begin{aligned} \text{Norms(per processorder)} \\ &= \text{Aggregated Consumption(per processorder)} \\ &\quad / \text{Aggregated Production (per processorder)} \end{aligned}$$

With the above logic applied on AOP period basis data we will get an array of data sets for each material for each grade per process order wise, now it is time to find final norm from the available data points. For specific to catalyst-chemicals logic used to calculate final norm is average out all norms after removing outlier to remove outlier logic used is explained below.

Outlier Removal:

Let's assume x, y, z are norms data set calculated with above explained logic.

Step 1 (A): Find average from all data points for each material (per grade).

Step 2 (B): Find standard deviation from all data points for each material (per grade).

Step 3(xi,yi,zi): calculate delta using norm data point average and standard deviation using below formula.

$$xi = Abs\left(\frac{x - A}{B}\right), yi = Abs\left(\frac{y - A}{B}\right), zi = Abs\left(\frac{z - A}{B}\right)$$

Step 4: removing data point (x,y,z...) (outlier removal) with condition remove norm data point if xi,yi,zi,...>1.

Step 5: average remaining norms data point which pass above condition.

The below screenshots depict the best-achieved feed stale-wise value captured in the AOP digitized platform

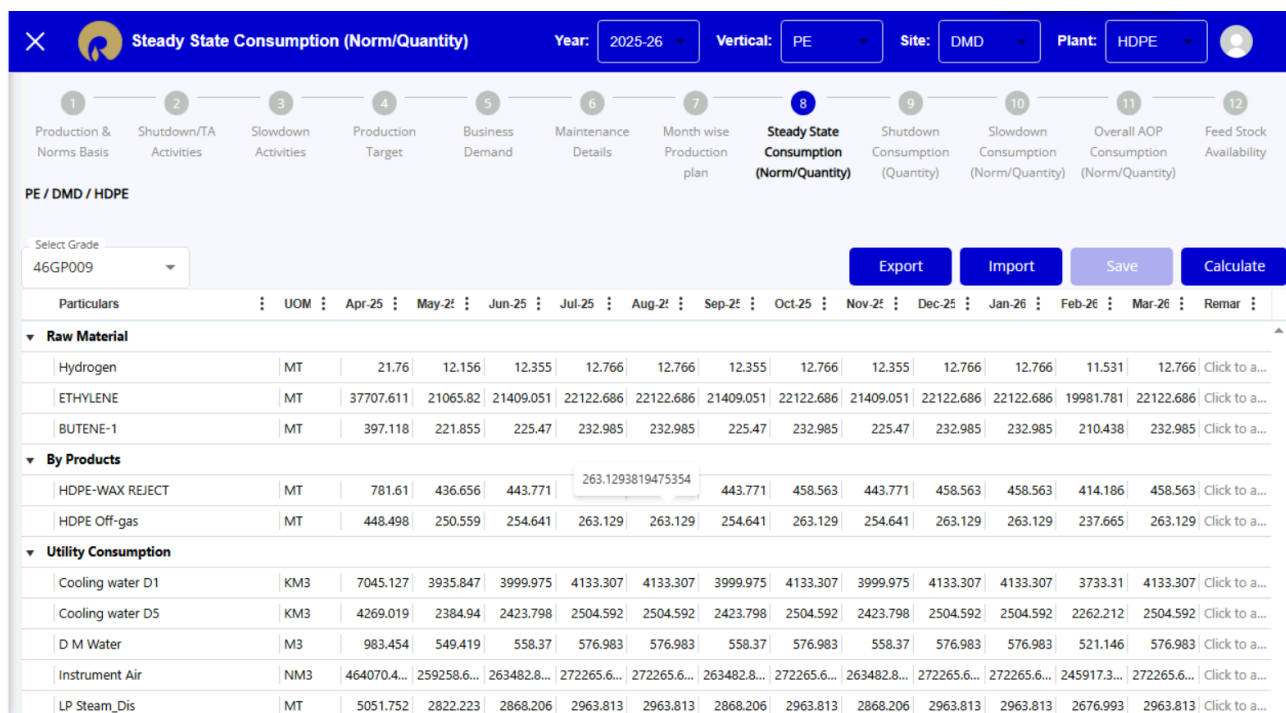


Figure 11 Digitized AOP screen for Best Achieved Norms

2.1.2.2 Shutdown & Slowdown Consumption.

Shutdown consumption is calculated by shutdown hours which is calculated based on shutdown schedule and shutdown consumption per hour which either constants or manual entry

$$\text{Shutdown Consumption} = \text{Number of shutdown hours} * \text{Shutdown consumption rate}$$

2.1.2.3 Budget Norms/ Consumption AOP

The shutdown consumption quantities will be summed with the steady-state operation norms, to derive the budgeted norms which will be inclusive of shutdown conditions. The intent here is to derive budgeted norms for the steady-state and shutdown months separately, using the logic below–

Consumption Norm

$$\begin{aligned} &= \text{Steady State Norms as captured in section 2.1.2.1} \\ &+ ((\text{Number of hours plant shutdown planned} * \text{Shutdown consumption per hour}) \\ &+ \text{Startuploss} + \text{Other losses}) / \text{Total Production} \end{aligned}$$

$$\text{Consumption AOP} = \text{Consumption Norm} * \text{Grade Production AOP}$$

If shutdown consumption is not applicable for any month, its value will be 0. Hence, the above formula is applicable for all the months.

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Select Grade

46GP009

Export

Import

Save

Calculate

Particulars	UOM	Apr-25	May-25	Jun-25	Jul-25	Aug-25	Sep-25	Oct-25	Nov-25	Dec-25	Jan-26	Feb-26	Mar-26	Remar
▼ Raw Material														
Hydrogen	MT	21.76	12.156	12.355	12.766	12.766	12.355	12.766	12.355	12.766	12.766	11.531	12.766	Click to a...
ETHYLENE	MT	37707.611	21065.82	21409.051	22122.686	22122.686	21409.051	22122.686	21409.051	22122.686	22122.686	19981.781	22122.686	Click to a...
BUTENE-1	MT	397.118	221.855	225.47	232.985	232.985	225.47	232.985	225.47	232.985	232.985	210.438	232.985	Click to a...
▼ By Products														
HDPE-WAX REJECT	MT	781.61	436.656	443.771	458.563	458.563	443.771	458.563	443.771	458.563	458.563	414.186	458.563	Click to a...
HDPE Off-gas	MT	448.498	250.559	254.641	263.129	263.129	254.641	263.129	254.641	263.129	263.129	237.665	263.129	Click to a...
▼ Utility Consumption														
Cooling water D1	KM3	7045.127	3935.847	3999.975	4133.307	4133.307	3999.975	4133.307	3999.975	4133.307	4133.307	3733.31	4133.307	Click to a...
Cooling water D5	KM3	4269.019	2384.94	2423.798	2504.592	2504.592	2423.798	2504.592	2423.798	2504.592	2504.592	2262.212	2504.592	Click to a...
D M Water	M3	983.454	549.419	558.37	576.983	576.983	558.37	576.983	558.37	576.983	576.983	521.146	576.983	Click to a...
Instrument Air	NM3	464070.4...	259258.6...	263482.8...	272265.6...	272265.6...	263482.8...	272265.6...	263482.8...	272265.6...	272265.6...	245917.3...	272265.6...	Click to a...

Figure 12 Overall AOP Consumption Screen

2.1.3 Feedstock Availability

This section to be updated after discussion in detail with RIL on Site wide feedstock/AOP Production screen.

3. AOP Reports

- Dataset (Basis)
- Annual AOP Cost.
- Plant Production Summary (T-14).
- Month wise Production Plan (T – 16)
- Month wise Raw data (T – 18).
- Turnaround Report (T – 19A).
- Annual Production Plan (T – 15).
- Plant Contribution (T – 21).

4. Key Assumptions and Exemptions

The above scope of work should be read with the following assumptions made

1. Data Acceptance
2. DCT provided by DMD Site team